MAE-263F Sample Midterm 1

Fall 2026

Wednesday, October 28, 2026

FULL NAME: _____

UID: _____

Time: 100 minutes (12:05 PM - 1:45 PM PDT)

Maximum points: 14.5 (29 questions × 0.5 pts)
Note: This sample midterm has 29 problems. The actual midterm will have 20 problems, each worth 0.5 pts, for a maximum of 10 points.

Please read the following instructions carefully:

1. This sample exam has 29 multiple-choice questions (the actual midterm has 20, each worth 0.5 pts). Each question is on its own page. For each question, **circle exactly ONE answer** (a, b, c, or d). **Only your a/b/c/d answer is graded.** Each question has space below it to show your work; **this work is NOT graded**.
2. You are allowed to use a calculator (including scientific and graphing) for mathematical operations, including finding roots of polynomials. Computers and internet access are NOT allowed.
3. This exam is open notes: you may bring ONE letter-sized two-sided cheat sheet and a printout of the course notes. You may write/type anything you wish on the cheat sheet.
4. Once you are done, please use your phone to **scan your midterm and upload it to Gradescope** before leaving the exam hall. Do NOT write anything once the exam is over at 1:45 pm. However, you may scan and upload after 1:45 pm.
5. Please make sure to **hand over the physical copy** of your midterm to the instructors.
6. If you are unable to scan your exam, please inform the instructor/TA and simply hand over the physical copy.

Honor Policy Statement

I agree to the UCLA student code of conduct for academic integrity. I affirm that I will not give or receive unauthorized help on this exam, and all work will be my own. I will not work on the exam outside the allowed time (12:05 pm - 1:45 pm PDT).

Signature: _____ Date: _____

Question 1 **0.5 pts**

Newton's method is used to solve $f(x) = x^2 - 3x + 2 = 0$ with the initial guess $x_0 = 0$. What is x_1 after **one** Newton iteration?

(a) $x_1 = 1$

(b) $x_1 = 2/3$

(c) $x_1 = -2/3$

(d) $x_1 = 3/2$

Space for work (not graded).

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Question 2 **0.5 pts**

Consider the system of equations

$$f_1 \equiv x_1^3 + x_2 = 0, \qquad f_2 \equiv -x_1 + x_2^3 + 1 = 0.$$

What is the Jacobian $\mathbf{J}$, with $J_{ij} = \partial f_i / \partial x_j$, evaluated at $(x_1, x_2) = (1, 2)$?

(a) $\begin{bmatrix} 3 & -1 \\ 1 & 12 \end{bmatrix}$

(b) $\begin{bmatrix} 3 & 1 \\ -1 & 6 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & 1 \\ -1 & 12 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 1 \\ -1 & 8 \end{bmatrix}$

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Question 3 **0.5 pts**

Consider the system of equations

$$f_1 \equiv x_1^2 - x_2 = 0, \qquad f_2 \equiv x_1 + x_2 - 2 = 0.$$

Starting from the initial guess $\mathbf{x}_0 = [0,\ 0]^T$, what is $\mathbf{x}_1$ after **one** Newton iteration, $\mathbf{x}_1 = \mathbf{x}_0 - \mathbf{J}^{-1}\mathbf{f}$?

(a) $\mathbf{x}_1 = [2,\ 0]^T$

(b) $\mathbf{x}_1 = [-2,\ 0]^T$

(c) $\mathbf{x}_1 = [1,\ 1]^T$

(d) $\mathbf{x}_1 = [0,\ 2]^T$

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Question 4 **0.5 pts**

For the same system as Question 3, $f_1 \equiv x_1^2 - x_2 = 0$ and $f_2 \equiv x_1 + x_2 - 2 = 0$, Newton's method fails at the first iteration if the Jacobian is singular ($\det \mathbf{J} = 0$). For which initial guess $\mathbf{x}_0 = [x_1,\ x_2]^T$ does this happen?

(a) $[0,\ 0]^T$

(b) $[1,\ 1]^T$

(c) $[0.5,\ -3]^T$

(d) $[-0.5,\ 3]^T$

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Question 5 **0.5 pts**

Two masses, each with $m = 1$ kg, move along the x-axis and are connected by one linear spring with stiffness $k = 50$ N/m. Neither mass is fixed, and there is no damping or external force. The elastic energy is $E = \frac{1}{2}k(x_2 - x_1 - L)^2$, where L is the rest length. The system is simulated with implicit Euler and $\Delta t = 0.1$ s, so the Jacobian is $J_{ij} = \frac{m_i}{\Delta t^2}\delta_{ij} + \frac{\partial^2 E}{\partial x_i \partial x_j}$. What is $\det \mathbf{J}$?

(a) 0

(b) 20,000

(c) 22,500

(d) 10,000

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Question 6 **0.5 pts**

For the same two-mass system as Question 5, the Hessian of the elastic energy is

$$\mathbf{H} = \frac{\partial^2 E}{\partial \mathbf{x} \partial \mathbf{x}} = k \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Which nonzero vector $\mathbf{v}$ lies in the null space of $\mathbf{H}$ (i.e., $\mathbf{H}\mathbf{v} = \mathbf{0}$)?

(a) $\mathbf{v} = [1,\ 1]^T$

(b) $\mathbf{v} = [1,\ -1]^T$

(c) $\mathbf{v} = [1,\ 0]^T$

(d) $\mathbf{v} = [0,\ 1]^T$

Space for work (not graded).

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Question 7 **0.5 pts**

What is the cross product $\mathbf{a} \times \mathbf{b}$, where $\mathbf{a} = [1,\ 1,\ 3]$ and $\mathbf{b} = [-2,\ 3,\ 5]$?

(a) $[4,\ 11,\ -5]$

(b) $[-4,\ 11,\ 5]$

(c) $[-4,\ -11,\ 5]$

(d) $[14,\ -11,\ 5]$

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Question 8 **0.5 pts**

A 2D discrete beam has three nodes, $\mathbf{x}_1 = (0, 0)$, $\mathbf{x}_2 = (1, 0)$, and $\mathbf{x}_3 = (1, -1)$. Using the turning angle $\phi_2 = \tan^{-1} \frac{(\mathbf{e}^1 \times \mathbf{e}^2) \cdot \hat{\mathbf{e}}_z}{\mathbf{e}^1 \cdot \mathbf{e}^2}$ and the discrete curvature $\kappa_2 = 2 \tan(\phi_2/2)$, what is κ_2? (Evaluate $\tan^{-1}$ as the four-quadrant arctangent, atan2(numerator, denominator).)

(a) 2

(b) -2

(c) 1

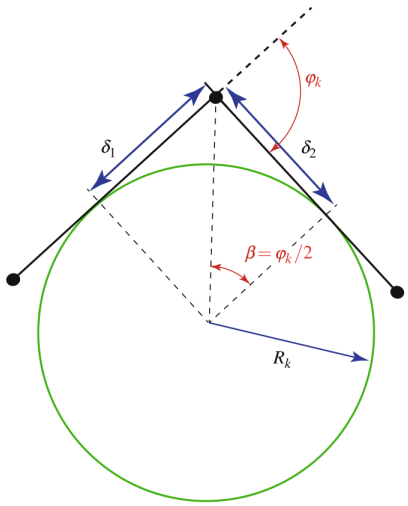
(d) $-\pi/2$

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Question 9 **0.5 pts**

The figure below shows two edges of a 2D discrete beam meeting at a node with turning angle φ_k. The circle (radius R_k) touches each edge at a distance $\delta_1 = \delta_2 = \delta$ from the node, and each radius drawn to a touching point is perpendicular to its edge.



If $\delta = 1$ m and $\varphi_k = 60°$, what is R_k?

(a) $1/\sqrt{3}$ m

(b) $2\sqrt{3}$ m

(c) 2 m

(d) $\sqrt{3}$ m

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Question 10 **0.5 pts**

A 2D discrete beam has three nodes, $\mathbf{x}_1 = (0, 0)$, $\mathbf{x}_2 = (2, 0)$, and $\mathbf{x}_3 = (2, 2)$, with all lengths in meters. Both edges have the same length, $\Delta l = 2$ m. What is the radius R_2 of the osculating circle at $\mathbf{x}_2$, i.e., the circle that touches each edge at its midpoint?

(a) 1 m

(b) 2 m

(c) 0.5 m

(d) 4 m

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Question 11 **0.5 pts**

A 2D discrete beam has three nodes, $\mathbf{x}_1 = (0, 0)$, $\mathbf{x}_2 = (1, 0)$, and $\mathbf{x}_3 = (1, 1)$, with all lengths in meters. The undeformed edge length is $\Delta l = 1$ m, the bending stiffness is $EI = 2$ N·m^2, and the natural curvature at $\mathbf{x}_2$ is $\kappa_2^0 = 1$. What is the bending energy $E_{b,2} = \frac{1}{2}\frac{EI}{\Delta l}(\kappa_2 - \kappa_2^0)^2$?

(a) 4 J

(b) 9 J

(c) 1 J

(d) 2 J

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Question 12 **0.5 pts**

An edge of a discrete beam connects $\mathbf{x}_1 = (0, 0)$ and $\mathbf{x}_2 = (3, 4)$, with lengths in meters. Its undeformed length is $\Delta l = 4$ m and its stretching stiffness is $EA = 100$ N. What is the stretching energy $E_s = \frac{1}{2} EA\, \varepsilon^2 \Delta l$, where $\varepsilon = \frac{|\mathbf{x}_2 - \mathbf{x}_1|}{\Delta l} - 1$?

(a) 50 J

(b) 3.125 J

(c) 25 J

(d) 12.5 J

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Question 13 **0.5 pts**

An edge of a discrete rod goes from $\mathbf{x}_1 = (0,0,0)$ to $\mathbf{x}_2 = (0,0,2)$. Which of the following $(\mathbf{m}_1, \mathbf{m}_2, \mathbf{t})$ is an orthonormal adapted frame on this edge? (Recall that it must satisfy $\mathbf{m}_1 \times \mathbf{m}_2 = \mathbf{t}$.)

(a) $\mathbf{m}_1 = (1,0,0)$, $\mathbf{m}_2 = (0,-1,0)$, $\mathbf{t} = (0,0,1)$

(b) $\mathbf{m}_1 = (0,1,0)$, $\mathbf{m}_2 = (-1,0,0)$, $\mathbf{t} = (0,0,1)$

(c) $\mathbf{m}_1 = (1,1,0)$, $\mathbf{m}_2 = (-1,1,0)$, $\mathbf{t} = (0,0,1)$

(d) $\mathbf{m}_1 = (1,0,0)$, $\mathbf{m}_2 = (0,0,-1)$, $\mathbf{t} = (0,1,0)$

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Question 14 **0.5 pts**

Why is a material frame (or, equivalently, a twist angle on each edge) needed in addition to the nodal coordinates to describe a 3D discrete elastic rod?

(a) The nodal coordinates describe only the centerline; they cannot capture the rotation (twist) of the cross-section about the centerline.

(b) The nodal coordinates cannot describe the bending curvature of a rod in 3D.

(c) The material frame is needed to compute the stretching energy of each edge.

(d) Without a material frame, the Jacobian of the 3D rod is always singular, even when some DOFs are fixed.

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Question 15 **0.5 pts**

A discrete rod has three nodes,

$$\mathbf{x}_0 = [0,\ 0,\ 0], \qquad \mathbf{x}_1 = [1,\ 0,\ 0], \qquad \mathbf{x}_2 = [1,\ 1,\ 0],$$

and two edges, $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{e}^1 = \mathbf{x}_2 - \mathbf{x}_1$. The vector $\mathbf{u} = [1,\ 0,\ 0]$ lies on the first edge. What is the result of parallel transporting $\mathbf{u}$ from the first edge ($\mathbf{e}^0$) to the second edge ($\mathbf{e}^1$)?

(a) $[1,\ 0,\ 0]$

(b) $[0,\ 0,\ 1]$

(c) $[0,\ 1,\ 0]$

(d) $[-1,\ 0,\ 0]$

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Question 16 **0.5 pts**

Which statement about parallel transport of a vector $\mathbf{u}$ (with $\mathbf{u} \perp \mathbf{t}^{k-1}$) from edge $\mathbf{e}^{k-1}$ to edge $\mathbf{e}^k$ is **FALSE**?

(a) The result is perpendicular to $\mathbf{t}^k$ and has the same length as $\mathbf{u}$.

(b) If $\mathbf{t}^k = \mathbf{t}^{k-1}$, the result is $\mathbf{u}$ itself.

(c) It is equivalent to rotating $\mathbf{u}$ about the axis $\mathbf{t}^{k-1} \times \mathbf{t}^k$ by the angle between $\mathbf{t}^{k-1}$ and $\mathbf{t}^k$.

(d) It rotates $\mathbf{u}$ about the tangent $\mathbf{t}^k$ by the integrated twist τ_k.

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Question 17 **0.5 pts**

Two consecutive edges of a rod are collinear, with $\mathbf{t}^1 = \mathbf{t}^2 = (1, 0, 0)$. The first material directors are $\mathbf{m}_1^1 = (0, 1, 0)$ and $\mathbf{m}_1^2 = (0, 0, 1)$. The integrated twist τ_2 is the signed angle from $P_1^2(\mathbf{m}_1^1)$ to $\mathbf{m}_1^2$ about $\mathbf{t}^2$. What is τ_2?

(a) $\pi/2$

(b) $-\pi/2$

(c) 0

(d) π

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Question 18 **0.5 pts**

In the DER algorithm (time-parallel reference frame), the twist angles on two consecutive edges are $\theta^1 = 0.1$ rad and $\theta^2 = 0.4$ rad, and the integrated reference twist at node $\mathbf{x}_2$ is $\Delta m_{2,\text{ref}} = -0.1$ rad. What is the integrated discrete twist τ_2?

(a) 0.4 rad

(b) 0.2 rad

(c) 0.3 rad

(d) 0.5 rad

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Question 19 **0.5 pts**

At node $\mathbf{x}_2$ of a discrete rod, the integrated twist is $\tau_2 = 0.5$ rad and the twisting stiffness is $GJ = 2$ N·m^2. The undeformed lengths of the two neighboring edges are $|\bar{\mathbf{e}}^1| = 1$ m and $|\bar{\mathbf{e}}^2| = 3$ m. What is the twisting energy $E_2^t = \frac{1}{2} GJ \tau_2^2 \frac{1}{l_2}$?

(a) 0.25 J

(b) 0.0625 J

(c) 0.125 J

(d) 0.5 J

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Question 20 **0.5 pts**

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [1,0,0]$, and $\mathbf{x}_2 = [1,1,0]$ (in meters), and two edges, $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{e}^1 = \mathbf{x}_2 - \mathbf{x}_1$. The material frames on the two edges are

$$\mathbf{m}_1^0 = [0,0,1],\ \mathbf{m}_2^0 = [0,-1,0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [1,0,0],\ \mathbf{m}_2^1 = [0,0,-1] \ \text{ on } \mathbf{e}^1.$$

The integrated twist τ_1 at node $\mathbf{x}_1$ is the signed angle from $P_0^1(\mathbf{m}_1^0)$ (the parallel transport of $\mathbf{m}_1^0$ from $\mathbf{e}^0$ to $\mathbf{e}^1$) to $\mathbf{m}_1^1$, about the axis $\mathbf{t}^1$. What is τ_1?

(a) 0

(b) $-\pi/2$

(c) π

(d) $\pi/2$

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Question 21 **0.5 pts**

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [1,0,0]$, and $\mathbf{x}_2 = [1,1,0]$ (in meters), and two edges, $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{e}^1 = \mathbf{x}_2 - \mathbf{x}_1$. The material frames on the two edges are

$$\mathbf{m}_1^0 = [0,0,1],\ \mathbf{m}_2^0 = [0,-1,0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [1,0,0],\ \mathbf{m}_2^1 = [0,0,-1] \ \text{ on } \mathbf{e}^1.$$

The rod is naturally untwisted, both edges have undeformed length 1 m, and the twisting stiffness is $GJ = 1$ N·m^2. What is the twisting energy $E_1^t = \frac{1}{2} GJ \tau_1^2 \frac{1}{\bar{l}_1}$ at node $\mathbf{x}_1$?

(a) $\pi^2/8$ J

(b) $\pi^2/4$ J

(c) $\pi/4$ J

(d) $\pi^2/16$ J

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Question 22 **0.5 pts**

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [1,0,0]$, and $\mathbf{x}_2 = [1,1,0]$ (in meters), and two edges, $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{e}^1 = \mathbf{x}_2 - \mathbf{x}_1$. The material frames on the two edges are

$$\mathbf{m}_1^0 = [0,0,1],\ \mathbf{m}_2^0 = [0,-1,0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [1,0,0],\ \mathbf{m}_2^1 = [0,0,-1] \ \text{ on } \mathbf{e}^1.$$

The (time-parallel) reference frames on the two edges are

$$\mathbf{a}_1^0 = [0,0,1],\ \mathbf{a}_2^0 = [0,-1,0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{a}_1^1 = \tfrac{1}{\sqrt{2}}[1,0,1],\ \mathbf{a}_2^1 = \tfrac{1}{\sqrt{2}}[1,0,-1] \ \text{ on } \mathbf{e}^1.$$

The reference twist $\Delta m_{1,\text{ref}}$ at node $\mathbf{x}_1$ is the signed angle from $P_0^1(\mathbf{a}_1^0)$ to $\mathbf{a}_1^1$, about the axis $\mathbf{t}^1$. What is $\Delta m_{1,\text{ref}}$?

(a) $-\pi/4$

(b) $\pi/4$

(c) $\pi/2$

(d) 0

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Question 23 **0.5 pts**

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [1,0,0]$, and $\mathbf{x}_2 = [1,1,0]$ (in meters), and two edges, $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{e}^1 = \mathbf{x}_2 - \mathbf{x}_1$. The material frames on the two edges are

$$\mathbf{m}_1^0 = [0,0,1],\ \mathbf{m}_2^0 = [0,-1,0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [1,0,0],\ \mathbf{m}_2^1 = [0,0,-1] \ \text{ on } \mathbf{e}^1.$$

The (time-parallel) reference frames on the two edges are

$$\mathbf{a}_1^0 = [0,0,1],\ \mathbf{a}_2^0 = [0,-1,0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{a}_1^1 = \tfrac{1}{\sqrt{2}}[1,0,1],\ \mathbf{a}_2^1 = \tfrac{1}{\sqrt{2}}[1,0,-1] \ \text{ on } \mathbf{e}^1.$$

In the DER algorithm, the twist angle θ^k (a degree of freedom) on edge $\mathbf{e}^k$ is the signed angle from $\mathbf{a}_1^k$ to $\mathbf{m}_1^k$ about $\mathbf{t}^k$. What are θ^0 and θ^1?

(a) $\theta^0 = \pi/2$, $\theta^1 = \pi/4$

(b) $\theta^0 = 0$, $\theta^1 = -\pi/4$

(c) $\theta^0 = 0$, $\theta^1 = \pi/4$

(d) $\theta^0 = 0$, $\theta^1 = \pi/2$

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Question 24 **0.5 pts**

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [1,0,0]$, and $\mathbf{x}_2 = [1,1,0]$ (in meters), and two edges, $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{e}^1 = \mathbf{x}_2 - \mathbf{x}_1$. The material frames on the two edges are

$$\mathbf{m}_1^0 = [0,0,1],\ \mathbf{m}_2^0 = [0,-1,0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [1,0,0],\ \mathbf{m}_2^1 = [0,0,-1] \ \text{ on } \mathbf{e}^1.$$

The (time-parallel) reference frames on the two edges are

$$\mathbf{a}_1^0 = [0,0,1],\ \mathbf{a}_2^0 = [0,-1,0] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{a}_1^1 = \tfrac{1}{\sqrt{2}}[1,0,1],\ \mathbf{a}_2^1 = \tfrac{1}{\sqrt{2}}[1,0,-1] \ \text{ on } \mathbf{e}^1.$$

Compute the integrated twist at node $\mathbf{x}_1$ a second way, using the DER formula $\tau_1 = (\theta^1 - \theta^0) + \Delta m_{1,\text{ref}}$. What is τ_1?

(a) $\pi/4$

(b) $\pi/2$

(c) 0

(d) $3\pi/4$

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Question 25 **0.5 pts**

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [2,0,0]$, and $\mathbf{x}_2 = [2,3,0]$ (in meters). Both edges have undeformed length 2 m, and the rod is naturally straight ($\boldsymbol{\kappa}^0 = \mathbf{0}$). The material frames are

$$\mathbf{m}_1^0 = [0,1,0],\ \mathbf{m}_2^0 = [0,0,1] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [-1,0,0],\ \mathbf{m}_2^1 = [0,0,1] \ \text{ on } \mathbf{e}^1.$$

The stiffnesses are $EA = 4$ N, $EI = 1$ N·m^2, and $GJ = 1$ N·m^2. What is the curvature binormal $(\kappa\mathbf{b})_1 = \dfrac{2\,\mathbf{t}^0 \times \mathbf{t}^1}{1 + \mathbf{t}^0 \cdot \mathbf{t}^1}$ at node $\mathbf{x}_1$?

(a) $[0,0,2]$

(b) $[0,0,-2]$

(c) $[0,0,1]$

(d) $[2,0,0]$

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Question 26 **0.5 pts**

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [2,0,0]$, and $\mathbf{x}_2 = [2,3,0]$ (in meters). Both edges have undeformed length 2 m, and the rod is naturally straight ($\boldsymbol{\kappa}^0 = \mathbf{0}$). The material frames are

$$\mathbf{m}_1^0 = [0,1,0],\ \mathbf{m}_2^0 = [0,0,1] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [-1,0,0],\ \mathbf{m}_2^1 = [0,0,1] \ \text{ on } \mathbf{e}^1.$$

The stiffnesses are $EA = 4$ N, $EI = 1$ N·m^2, and $GJ = 1$ N·m^2. The material curvatures at node $\mathbf{x}_1$ are $\kappa_1 = \frac{1}{2}(\kappa\mathbf{b})_1 \cdot (\mathbf{m}_2^0 + \mathbf{m}_2^1)$ and $\kappa_2 = -\frac{1}{2}(\kappa\mathbf{b})_1 \cdot (\mathbf{m}_1^0 + \mathbf{m}_1^1)$. What is $\boldsymbol{\kappa}_1 = [\kappa_1,\ \kappa_2]$?

(a) $[0,\ 2]$

(b) $[0,\ -2]$

(c) $[2,\ 0]$

(d) $[1,\ 0]$

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Question 27 **0.5 pts**

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [2,0,0]$, and $\mathbf{x}_2 = [2,3,0]$ (in meters). Both edges have undeformed length 2 m, and the rod is naturally straight ($\boldsymbol{\kappa}^0 = \mathbf{0}$). The material frames are

$$\mathbf{m}_1^0 = [0,1,0],\ \mathbf{m}_2^0 = [0,0,1] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [-1,0,0],\ \mathbf{m}_2^1 = [0,0,1] \ \text{ on } \mathbf{e}^1.$$

The stiffnesses are $EA = 4$ N, $EI = 1$ N·m^2, and $GJ = 1$ N·m^2. What is the bending energy $E_1^b = \frac{1}{2} EI\, |\boldsymbol{\kappa}_1 - \boldsymbol{\kappa}_1^0|^2 \, \frac{1}{\bar{l}_1}$ at node $\mathbf{x}_1$?

(a) 2 J

(b) 0.5 J

(c) 4 J

(d) 1 J

Space for work (not graded).

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Question 28 **0.5 pts**

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [2,0,0]$, and $\mathbf{x}_2 = [2,3,0]$ (in meters). Both edges have undeformed length 2 m, and the rod is naturally straight ($\boldsymbol{\kappa}^0 = \mathbf{0}$). The material frames are

$$\mathbf{m}_1^0 = [0,1,0],\ \mathbf{m}_2^0 = [0,0,1] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [-1,0,0],\ \mathbf{m}_2^1 = [0,0,1] \ \text{ on } \mathbf{e}^1.$$

The stiffnesses are $EA = 4$ N, $EI = 1$ N·m^2, and $GJ = 1$ N·m^2. What is the total stretching energy of the rod, $E^s = \sum_k \frac{1}{2} EA \left(\frac{|\mathbf{e}^k|}{|\bar{\mathbf{e}}^k|} - 1 \right)^2 |\bar{\mathbf{e}}^k|$?

(a) 0.5 J

(b) 2 J

(c) 4 J

(d) 1 J

Space for work (not graded).

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Question 29 **0.5 pts**

A discrete rod has three nodes, $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [2,0,0]$, and $\mathbf{x}_2 = [2,3,0]$ (in meters). Both edges have undeformed length 2 m, and the rod is naturally straight ($\boldsymbol{\kappa}^0 = \mathbf{0}$). The material frames are

$$\mathbf{m}_1^0 = [0,1,0],\ \mathbf{m}_2^0 = [0,0,1] \ \text{ on } \mathbf{e}^0, \qquad \mathbf{m}_1^1 = [-1,0,0],\ \mathbf{m}_2^1 = [0,0,1] \ \text{ on } \mathbf{e}^1.$$

The stiffnesses are $EA = 4$ N, $EI = 1$ N·m^2, and $GJ = 1$ N·m^2. The rod is naturally untwisted. What is the total elastic energy (stretching + bending + twisting) of the rod?

(a) 1 J

(b) 2 J

(c) 3 J

(d) 0 J

Space for work (not graded).

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